

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
```

```
df=pd.read_csv(r"/content/sample_data/world_population.csv")
df
```

	Rank	CCA3	Country	Capital	Continent	2022 Population	2020 Population	2015 Population	2010 Population
0	36	AFG	Afghanistan	Kabul	Asia	41128771.0	38972230.0	33753499.0	28189672.0
1	138	ALB	Albania	Tirana	Europe	2842321.0	2866849.0	2882481.0	2913399.0
2	34	DZA	Algeria	Algiers	Africa	44903225.0	43451666.0	39543154.0	35856344.0
3	213	ASM	American Samoa	Pago Pago	Oceania	44273.0	46189.0	51368.0	54849.0
4	203	AND	Andorra	Andorra la Vella	Europe	79824.0	77700.0	71746.0	71519.0
...	...	...	...	...	...	...	...	...	...
229	226	WLF	Wallis and Futuna	Mata-Utu	Oceania	11572.0	11655.0	12182.0	13142.0
230	172	ESH	Western Sahara	El Aaiún	Africa	575986.0	556048.0	491824.0	413296.0
231	46	YEM	Yemen	Sanaa	Asia	33696614.0	32284046.0	28516545.0	24743946.0
232	63	ZMB	Zambia	Lusaka	Africa	20017675.0	18927715.0	NaN	13792086.0
233	74	ZWE	Zimbabwe	Harare	Africa	16320537.0	15669666.0	14154937.0	12839771.0

234 rows × 17 columns

Next steps: [Generate code with df](#) [New interactive sheet](#)

```
pd.set_option('display.float_format', lambda x: '%.2f' % x)
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 234 entries, 0 to 233
Data columns (total 17 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Rank                                  234 non-null    int64
1   CCA3                                  234 non-null    object
2   Country                              234 non-null    object
3   Capital                              234 non-null    object
4   Continent                            234 non-null    object
5   2022 Population                      230 non-null    float64
6   2020 Population                      233 non-null    float64
7   2015 Population                      230 non-null    float64
8   2010 Population                      227 non-null    float64
```

```

9   2000 Population      227 non-null    float64
10  1990 Population      229 non-null    float64
11  1980 Population      229 non-null    float64
12  1970 Population      230 non-null    float64
13  Area (km²)           232 non-null    float64
14  Density (per km²)    230 non-null    float64
15  Growth Rate          232 non-null    float64
16  World Population Percentage 234 non-null    float64
dtypes: float64(12), int64(1), object(4)
memory usage: 31.2+ KB

```

```

# Shape and structure
print(df.shape)      # rows, columns

```

```

(234, 17)

```

```

print(df.info())      # column types, null counts

```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 234 entries, 0 to 233
Data columns (total 17 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Rank                                  234 non-null    int64
1   CCA3                                  234 non-null    object
2   Country                              234 non-null    object
3   Capital                              234 non-null    object
4   Continent                             234 non-null    object
5   2022 Population                       230 non-null    float64
6   2020 Population                       233 non-null    float64
7   2015 Population                       230 non-null    float64
8   2010 Population                       227 non-null    float64
9   2000 Population                       227 non-null    float64
10  1990 Population                       229 non-null    float64
11  1980 Population                       229 non-null    float64
12  1970 Population                       230 non-null    float64
13  Area (km²)                            232 non-null    float64
14  Density (per km²)                     230 non-null    float64
15  Growth Rate                           232 non-null    float64
16  World Population Percentage           234 non-null    float64
dtypes: float64(12), int64(1), object(4)
memory usage: 31.2+ KB
None

```

```

print(df.head())      # first 5 rows

```

```

      Rank CCA3      Country      Capital Continent  2022 Population \
0      36  AFG      Afghanistan      Kabul      Asia    41128771.00
1     138  ALB      Albania      Tirana      Europe    2842321.00
2      34  DZA      Algeria      Algiers      Africa    44903225.00
3     213  ASM  American Samoa      Pago Pago  Oceania      44273.00
4     203  AND      Andorra  Andorra la Vella  Europe     79824.00

      2020 Population  2015 Population  2010 Population  2000 Population \
0    38972230.00    33753499.00    28189672.00    19542982.00
1     2866849.00     2882481.00     2913399.00     3182021.00
2   43451666.00   39543154.00   35856344.00   30774621.00
3     46189.00     51368.00     54849.00     58230.00
4     77700.00     71746.00     71519.00     66097.00

```

	1990 Population	1980 Population	1970 Population	Area (km ²) \
0	10694796.00	12486631.00	10752971.00	652230.00
1	3295066.00	2941651.00	2324731.00	28748.00
2	25518074.00	18739378.00	13795915.00	2381741.00
3	47818.00	32886.00	27075.00	199.00
4	53569.00	35611.00	19860.00	468.00

	Density (per km ²)	Growth Rate	World Population Percentage
0	63.06	1.03	0.52
1	98.87	1.00	0.04
2	18.85	1.02	0.56
3	222.48	0.98	0.00
4	170.56	1.01	0.00

```
# Numeric features
print(df.describe())
```

	Rank	2022 Population	2020 Population	2015 Population \
count	234.00	230.00	233.00	230.00
mean	117.50	34632250.88	33600710.95	32066004.16
std	67.69	137889172.44	135873196.61	131507146.34
min	1.00	510.00	520.00	564.00
25%	59.25	419738.50	406471.00	394295.00
50%	117.50	5762857.00	5456681.00	5244415.00
75%	175.75	22653719.00	21522626.00	19730853.75
max	234.00	1425887337.00	1424929781.00	1393715448.00

	2010 Population	2000 Population	1990 Population	1980 Population \
count	227.00	227.00	229.00	229.00
mean	30270164.48	26840495.26	19330463.93	16282884.78
std	126074183.54	113352454.57	81309624.96	69345465.54
min	596.00	651.00	700.00	733.00
25%	382726.50	329470.00	261928.00	223752.00
50%	4889741.00	4491202.00	3785847.00	3135123.00
75%	16825852.50	15625467.00	11882762.00	9817257.00
max	1348191368.00	1264099069.00	1153704252.00	982372466.00

	1970 Population	Area (km ²)	Density (per km ²)	Growth Rate \
count	230.00	232.00	230.00	232.00
mean	15866499.13	581663.75	456.81	1.01
std	68355859.75	1769133.06	2083.74	0.01
min	752.00	1.00	0.03	0.91
25%	145880.50	2567.25	36.60	1.00
50%	2511718.00	77141.00	95.35	1.01
75%	8817329.00	414643.25	236.88	1.02
max	822534450.00	17098242.00	23172.27	1.07

	World Population Percentage
count	234.00
mean	0.43
std	1.71
min	0.00
25%	0.01
50%	0.07
75%	0.28
max	17.88

```
# Categorical features
print(df.describe(include='object'))
```

	CCA3	Country	Capital	Continent
count	234	234	234	234
unique	234	234	234	6
top	AFG	Afghanistan	Kabul	Africa
freq	1	1	1	57

df.sort_values(by="World Population Percentage", ascending=False).head(10)								
	Rank	CCA3	Country	Capital	Continent	2022 Population	2020 Population	2015 Population
41	1	CHN	China	Beijing	Asia	1425887337.00	1424929781.00	1393715448.00
92	2	IND	India	New Delhi	Asia	1417173173.00	1396387127.00	1322866505.00
221	3	USA	United States	Washington, D.C.	North America	338289857.00	335942003.00	324607776.00
93	4	IDN	Indonesia	Jakarta	Asia	275501339.00	271857970.00	259091970.00
156	5	PAK	Pakistan	Islamabad	Asia	235824862.00	227196741.00	210969298.00
149	6	NGA	Nigeria	Abuja	Africa	218541212.00	208327405.00	183995785.00
27	7	BRA	Brazil	Brasilia	South America	215313498.00	213196304.00	205188205.00
16	8	BGD	Bangladesh	Dhaka	Asia	171186372.00	167420951.00	157830000.00
171	9	RUS	Russia	Moscow	Europe	144713314.00	145617329.00	144668389.00
131	10	MEX	Mexico	Mexico City	North America	127504125.00	125998302.00	120149897.00

# Missing values print(df.isnull().sum())	
Rank	0
CCA3	0
Country	0
Capital	0
Continent	0
2022 Population	4
2020 Population	1
2015 Population	4
2010 Population	7
2000 Population	7
1990 Population	5
1980 Population	5
1970 Population	4
Area (km²)	2
Density (per km²)	4
Growth Rate	2
World Population Percentage	0
dtype: int64	

```
# Duplicate rows
print(df.duplicated().sum())
```

```
0
```

```
# Unique values per column
print(df.nunique())
```

```
Rank                234
CCA3                234
Country            234
Capital            234
Continent             6
2022 Population    230
2020 Population    233
2015 Population    230
2010 Population    227
2000 Population    227
1990 Population    229
1980 Population    229
1970 Population    230
Area (km²)         231
Density (per km²)  230
Growth Rate        178
World Population Percentage  70
dtype: int64
```

```
df.corr(numeric_only=True)
```

	Rank	2022 Population	2020 Population	2015 Population	2010 Population	2000 Population	1990 Population	Pop
Rank	1.00	-0.36	-0.36	-0.35	-0.35	-0.34	-0.33	
2022 Population	-0.36	1.00	1.00	1.00	1.00	0.99	0.99	
2020 Population	-0.36	1.00	1.00	1.00	1.00	1.00	0.99	
2015 Population	-0.35	1.00	1.00	1.00	1.00	1.00	0.99	
2010 Population	-0.35	1.00	1.00	1.00	1.00	1.00	1.00	
2000 Population	-0.34	0.99	1.00	1.00	1.00	1.00	1.00	
1990 Population	-0.33	0.99	0.99	0.99	1.00	1.00	1.00	
1980 Population	-0.33	0.99	0.99	0.99	0.99	1.00	1.00	
1970 Population	-0.34	0.97	0.98	0.98	0.98	0.99	1.00	
Area (km²)	-0.38	0.45	0.45	0.46	0.46	0.47	0.52	
Density (per km²)	0.13	-0.03	-0.03	-0.03	-0.03	-0.03	-0.03	
Growth Rate	-0.22	-0.02	-0.03	-0.03	-0.04	-0.05	-0.07	
World Population Percentage	-0.36	1.00	1.00	1.00	1.00	0.99	0.99	

```
sns.heatmap(df.corr(numeric_only=True), annot=True)
plt.rcParams['figure.figsize']=(20,7)
plt.show()
```



```
df.groupby('Continent').mean(numeric_only=True).sort_values(by="2022 Population", ascending=True)
```

	Rank	2022 Population	2020 Population	2015 Population	2010 Population	2000 Population	1990 Population
Continent							
Asia	77.56	96327387.31	94955134.37	89165003.64	89087770.00	80580835.11	48639995.33
South America	97.57	31201186.29	30823574.50	29509599.71	26789395.54	25015888.69	21224743.93
Africa	92.16	25455879.68	23871435.26	21419703.57	18898197.31	14598365.95	11376964.52
Europe	124.50	15055371.82	14915843.92	15027454.12	14712278.68	14817685.71	14785203.94
North America	160.93	15007403.40	14855914.82	14259596.25	13568016.28	12151739.60	10531660.62
Oceania	188.52	2046386.32	1910148.96	1756664.48	1613163.65	1357512.09	1162774.87

```
df[df['Continent'].str.contains('Oceania')]
```

Rank	CCA3	Country	Capital	Continent	2022 Population	2020 Population	2015 Population	Popul
213	ASM	American Samoa	Pago Pago	Oceania	44273.00	46189.00	51368.00	54
55	AUS	Australia	Canberra	Oceania	26177413.00	25670051.00	23820236.00	22019
223	COK	Cook Islands	Avarua	Oceania	17011.00	17029.00	17695.00	17
162	FJI	Fiji	Suva	Oceania	929766.00	920422.00	917200.00	905
183	PYF	French Polynesia	Papeete	Oceania	306279.00	301920.00	291787.00	283
191	GUM	Guam	Hagåtña	Oceania	171774.00	169231.00	167978.00	164
192	KIR	Kiribati	Tarawa	Oceania	131232.00	126463.00	116707.00	107
215	MHL	Marshall Islands	Majuro	Oceania	41569.00	43413.00	49410.00	53
194	FSM	Micronesia	Palikir	Oceania	114164.00	112106.00	109462.00	107
225	NRU	Nauru	Yaren	Oceania	12668.00	12315.00	11185.00	10
185	NCL	New Caledonia	Nouméa	Oceania	289950.00	286403.00	283032.00	261
123	NZL	New Zealand	Wellington	Oceania	5185288.00	5061133.00	4590590.00	4346
232	NIU	Niue	Alofi	Oceania	1934.00	1942.00	1847.00	1
210	NFK	Northern Mariana Islands	Saipan	Oceania	49551.00	49587.00	51514.00	54
222	PLW	Palau	Ngerulmud	Oceania	NaN	17972.00	17794.00	18
93	PNG	Papua New Guinea	Port Moresby	Oceania	10142619.00	9749640.00	8682174.00	7583
188	WSM	Samoa	Apia	Oceania	222382.00	214929.00	203571.00	194
166	SLB	Solomon Islands	Honiara	Oceania	724273.00	691191.00	612660.00	540
233	TKL	Tokelau	Nukunonu	Oceania	1871.00	1827.00	1454.00	1
197	TON	Tonga	Nuku'alofa	Oceania	106858.00	105254.00	106122.00	107
227	TUV	Tuvalu	Funafuti	Oceania	11312.00	11069.00	10877.00	10
181	VUT	Vanuatu	Port-Vila	Oceania	326740.00	311685.00	276438.00	245
226	WLF	Wallis and Futuna	Mata-Utu	Oceania	11572.00	11655.00	12182.00	13

```
df2=df.groupby('Continent')[['1970 Population', '1980 Population','1990 Population',
'2010 Population', '2015 Population', '2020 Population',
```



```
'2022 Population']].mean().sort_values(by="2022 Population", ascending=False)
df2
```

	1970 Population	1980 Population	1990 Population	2000 Population	2010 Population	2015 Population	Popul
Continent							
Asia	43839877.83	40278333.33	48639995.33	80580835.11	89087770.00	89165003.64	949551
South America	13781939.71	17270643.29	21224743.93	25015888.69	26789395.54	29509599.71	308235
Africa	6567175.27	8586031.98	11376964.52	14598365.95	18898197.31	21419703.57	238714
Europe	13118479.82	14200004.52	14785203.94	14817685.71	14712278.68	15027454.12	149158
North America	7885865.15	9207334.03	10531660.62	12151739.60	13568016.28	14259596.25	148559
Oceania	846968.26	996532.17	1162774.87	1357512.09	1613163.65	1756664.48	19107

Next steps:

[Generate code with df2](#)

[New interactive sheet](#)

df.columns

```
Index(['Rank', 'CCA3', 'Country', 'Capital', 'Continent', '2022 Population',
      '2020 Population', '2015 Population', '2010 Population',
      '2000 Population', '1990 Population', '1980 Population',
      '1970 Population', 'Area (km²)', 'Density (per km²)', 'Growth Rate',
      'World Population Percentage'],
      dtype='object')
```

```
df3=df2.transpose()
```

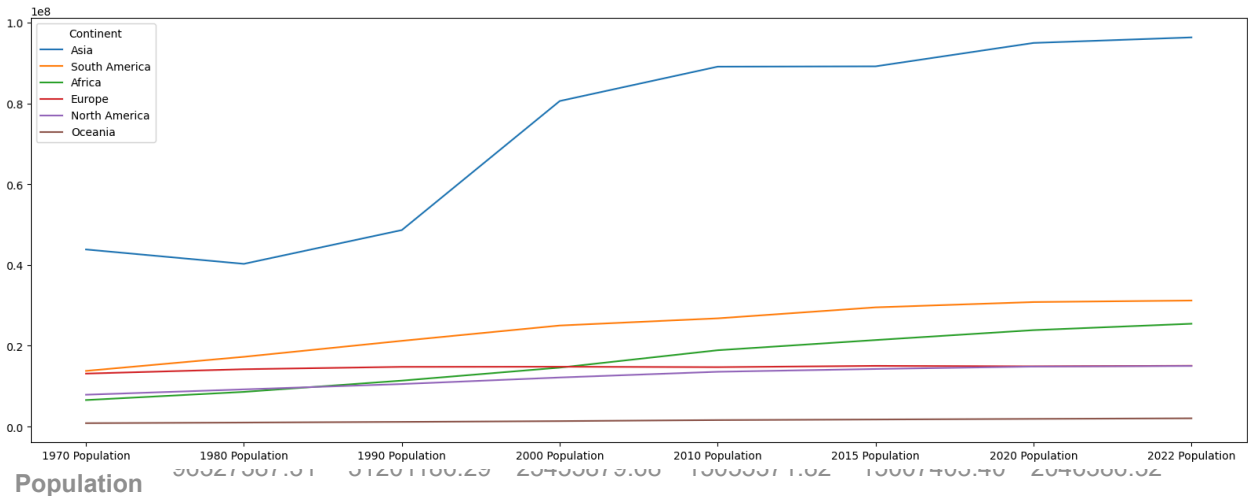
df3

Continent	Asia	South America	Africa	Europe	North America	Oceania
1970	43839877.83	13781939.71	6567175.27	13118479.82	7885865.15	846968.26

Next steps: [Generate code with df3](#) [New interactive sheet](#)

df3.plot()

Population
<Axes: >



df.boxplot(figsize=(20,10), grid=False)

<Axes: >

